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PERFORMANCE SPECIFICATION

GREASE, HIGH PERFORMANCE, MULTIPURPOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of high performance grease for multipurpose use and use in lubricated ball and roller bearings operating continuously in a temperature range of -29 to 149 °C (-20 to 300 °F) and intermittently up to 177 °C (350 °F) for periods up to 4 hours in any 24-hour period.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791 - Testing Method of Lubricants, Liquid Fuels, and Related Products

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-B-17931 - Bearings, Ball, Annular, for Quiet Operation

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual

(Copies of the chapter titled "Material Control Program" are available by email request to CommandStandards@navy.mil.)

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering [hard copy] via the Naval Logistics Library [NLL] at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B150/B150M - Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

ASTM D217 - Standard Test Methods for Cone Penetration of Lubricating Grease

ASTM D942 - Standard Test Method for Oxidation Stability of Lubricating Greases by the Oxygen Pressure Vessel Method

ASTM D1264 - Standard Test Method for Determining the Water Washout Characteristics of Lubricating Greases

ASTM D1478 - Standard Test Method for Low-Temperature Torque of Ball Bearing Grease

ASTM D1743 - Standard Test Method for Determining Corrosion Preventive Properties of Lubricating Greases

ASTM D2265 - Standard Test Method for Dropping Point of Lubricating Grease Over Wide Temperature Range

ASTM D2266 - Standard Test Method for Wear Preventive Characteristics of Lubricating Grease (Four-Ball Method)

ASTM D2595 - Standard Test Method for Evaporation Loss of Lubricating Greases Over Wide-Temperature Range

ASTM D2596 - Standard Test Method for Measurement of Extreme-Pressure Properties of Lubricating Grease (Four-Ball Method)

ASTM D3336 - Standard Test Method for Life of Lubricating Greases in Ball Bearings at Elevated Temperatures

ASTM D3704 - Standard Test Method for Wear Preventive Properties of Lubricating Greases Using the (Falex) Block on Ring Test Machine in Oscillating Motion

ASTM D4048 - Standard Test Method for Detection of Copper Corrosion from Lubricating Grease

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM D4289 - Standard Test Method for Elastomer Compatibility of Lubricating Greases and Fluids

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ASTM D6184	-	Standard Test Method for Oil Separation from Lubricating Grease (Conical Sieve Method)
ASTM E29	-	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. For purposes of determining conformance with each requirement, an observed value or calculated value shall be rounded off to the nearest unit in the last right-hand place of figures used in expressing the limiting value in accordance with the rounding method in ASTM E29.

3.2 Qualification. The grease furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.3 Material. The grease shall consist of a liquid lubricant and gelling agent that when formulated results in a compound meeting this specification. The grease shall not be comprised of silicones or contain silicone additives.

3.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Toxicity and prohibited materials.

3.5.1 Toxicity. When evaluated in accordance with 4.5.1, the grease shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5.1 and 6.6).

3.5.2 Prohibited materials. The grease shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 (see 4.5.2).

3.6 Off-gassing. The grease shall be evaluated for off-gassing in accordance with the requirements of 4.5.3. Based on the circumstances of use and the chemical nature of the grease, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the grease shall be assigned to either the “Permitted” or “Limited” category (see 4.5.3 and 6.5).

3.7 Ozone depleting substances (ODS). The use of ODSs in the composition of the grease under this specification directly or referenced in any federal test method is hereto prohibited. Environmentally safe and non-ODS alternative solvents shall be substituted in test methods for any previously specified ODS such as 1,1,1-trichloroethane, tri-chlorofluoroethane, perchloroethane, or other chlorofluorocarbons. A dehydrated mixture of primary alcohol, such as ethanol or isopropyl alcohol, and a non-aromatic hydrocarbon solvent resulting in a minimum flash point of 60.0 °C (140.0 °F) plus effective solvent action shall be used as an alternative for any ODS.

3.8 Odor. The grease shall not have an objectionable odor, or odor of rancidity, perfume or free alcohol (see 4.5.5).

3.9 Workmanship. The grease, when examined visually, shall be a smooth and homogenous mixture, free from lumps and extraneous materials (see 4.5.6).

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3.10 Compatibility. The grease shall be compatible with all previously qualified greases when tested in accordance with 4.5.7. Immediately after mixing, and after undisturbed storage for 7 days, the mixture shall not form resinous gums, sludge, or insoluble solid materials and shall comply with the worked penetration and penetration after prolonged working requirements (see table I). In the event of failure, the previously qualified grease (unmixed with other greases) shall be tested to verify that it meets the worked penetration and worked stability requirements under the conditions specified in 4.5.7. If the previously qualified sample fails to meet the worked penetration or worked stability requirement, a fresh sample shall be obtained and the compatibility test repeated.

3.11 Chemical and physical properties. The grease shall have the chemical and physical properties specified in table I.

TABLE I. Chemical and physical requirements and test methods.

Property	Requirement	Test Method (ASTM or Other) or Verification Paragraph
Dropping point, °C (°F), min	232 (450)	ASTM D2265
Penetration, worked (60 double strokes)	265-320	ASTM D217
Penetration of lubricating greases after prolonged working (after 100,000 double strokes), max 1/	350	ASTM D217
Evaporation after 22 hours at 177±3 ° C (350±5 °F), % max	12.0	ASTM D2595
Oil separation after 30 hours at 177±3 °C (350±5 °F), % max	10.0	ASTM D6184
Dirt content, particles per milliliter grease, max:		FED-STD-791, method 3005
25-74 µm 2/	1000	
75 µm or larger	0	
Corrosiveness (copper strip), rating, max	1b	ASTM D4048
Oxidation stability, pressure drop, kilopascals (pounds per square inch), max:		ASTM D942
After 100 hours	103 (15)	
After 500 hours	172 (25)	
Water washout at 38 °C (100 °F), % max	10.0	ASTM D1264
Load wear index (kgf), min	30	ASTM D2596
Low temperature torque at -29±1 °C (-20±2 °F), newton-meter (grams-centimeter), max:		ASTM D1478
Start	0.441 (4500)	
Running, after 1 hour	0.147 (1500)	
Steel on steel wear, millimeter (mm), max	1.30 average diameter of wear scars on the three steel balls	ASTM D2266, option A
Rust preventive properties, rating	Pass	ASTM D1743
High temperature performance at 177 °C (350 °F), hours, min	400	ASTM D3336
Swelling of synthetic rubber, NBR-L, after 168±0.5 hours at 70±1 °C (158±2 °F), % max	10.0	ASTM D4289

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TABLE I. Chemical and physical requirements and test methods – Continued.

Gear wear (milligrams/1000 cycles), max:		FED-STD-791, method 335, brass gear
Under 2.3 kilograms (kg) (5 pounds [lb]) load	2.5	
Under 4.5 kg (10 lb) load	3.5	
Oscillation friction:		ASTM D3704
Wear, mm, max	6.35	
Coefficient of friction, max [3/]	0.44	
Storage stability, penetration:		4.5.8.1
Unworked, min	200	
Worked (compared to original), max	±30	
Separation of crystalline material (6 months)	None	
Performance life test [4/]	Performance comparable or superior to Navy Reference Grease Z-1 (see 6.7)	4.5.8.2

NOTES:

[1/] Test may be referred to as "worked stability" in some publications.

[2/] Micrometer (μm)

[3/] Time: 35,000 cycles; angle of oscillation: 90 degrees; oscillation speed: 87.5 cycles per minute; test ring: type S-10; test block shall be an ASTM B150/B150M, alloy C64200 having a Rockwell B hardness of HRB 88±3 with dimensions and surface finish the same as those specified for H-30 or H-60 blocks specified in the ASTM method; loading: 30 lbf at 60 cycles and 60 lbf at 120 cycles.

[4/] Information about approved test facilities can be obtained by contacting NAVSEA (see 6.7).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the conditions specified in table I on unworked grease. The physical values specified in table I apply to the average of determinations made on the sample.

4.3 Qualification inspection. Qualification tests shall be conducted at a laboratory acceptable to NAVSEA. Qualification inspection shall consist of all the tests specified in the qualification test column of table II to satisfy the requirements in section 3 and table I (see 6.3).

4.3.1 Change approval. A change in product composition, formulation, or ingredients used in the manufacture of grease that has been qualified shall require written approval of NAVSEA. Incorporation of any changes may require requalification of the item in question.

4.3.2 Requalification. Unless otherwise approved by NAVSEA, requalification shall consist of all the tests specified in the qualification test column of table II to satisfy the requirements in section 3 and table I (see 6.3).

4.4 Conformance inspection. Conformance inspection shall consist of all the tests specified in the conformance test column of table II. The inspection shall be conducted on samples taken in accordance with 4.4.1.

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4.4.1 Sampling for tests. Two 2.95-kg (6.5-lb) containers of grease shall be selected, taken at random from each filled lot (see 6.8.1). For users who obtain grease in large containers, samples shall be selected in accordance with ASTM D4057. One sample shall be used for tests and the other shall be retained. If any test result does not comply with the stated requirement, the test in question shall be repeated on the retained sample. If the sample fails again, the lot shall be rejected.

TABLE II. Qualification tests and conformance tests.

Tests	Requirement	Verification	Qualification Test	Conformance Test
Toxicity	3.5.1	4.5.1	X	
Prohibited materials	3.5.2	4.5.2	X	
Off-gassing	3.6	4.5.3	X	
ODSs	3.7	4.5.4	X	
Odor	3.8	4.5.5	X	X
Workmanship	3.9	4.5.6	X	X
Compatibility	3.10	4.5.7	X	
Dropping point	3.11	4.5.8	X	X
Penetration, worked	3.11	4.5.8	X	X
Penetration, after prolonged working	3.11	4.5.8	X	
Evaporation	3.11	4.5.8	X	X
Oil separation	3.11	4.5.8	X	X
Dirt content	3.11	4.5.8	X	X
Corrosiveness (copper strip)	3.11	4.5.8	X	X
Oxidation stability	3.11	4.5.8	X	
Water washout	3.11	4.5.8	X	
Load wear index	3.11	4.5.8	X	
Low temperature torque	3.11	4.5.8	X	X
Steel on steel wear	3.11	4.5.8	X	X
Rust preventive properties	3.11	4.5.8	X	
High temperature performance	3.11	4.5.8	X	
Swelling of synthetic rubber	3.11	4.5.8	X	
Gear wear (mg/1000 cycles)	3.11	4.5.8	X	
Oscillation friction	3.11	4.5.8	X	
Storage stability	3.11	4.5.8.1	X	
Performance life test	3.11	4.5.8.2	X	

4.5 Test methods.

4.5.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.5.1 as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the grease using data provided by the manufacturer/distributor to the NMCPHC (see 3.5.1 and 6.6).

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4.5.2 Prohibited materials. Prohibited materials shall be verified for conformance to 3.5.2 as required by the qualifying activity.

4.5.3 Off-gassing. The grease shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled "Material Control Program" (see 3.6 and 6.5). If the Navy determines that off-gas testing is required, testing shall be conducted at a NAVSEA approved test facility (see 6.5). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.6).

4.5.4 ODSs. Verification that an ODS is not used in the formulation of the grease or testing thereof shall be accomplished as specified by the qualifying activity (see 3.7).

4.5.5 Odor. The odor of the grease shall be observed immediately upon opening the test container (see 3.8).

4.5.6 Visual examination. Visual examination shall be conducted to determine conformance with the requirements of 3.9.

4.5.7 Compatibility. Standard grease cups shall be filled with equal volumes of the grease being tested and previously qualified greases. The cup shall be filled by alternately adding 60 grams of each of the two greases being tested, until the cup is full (approximately 454 grams). The two greases shall be mixed by working the sample for 60 double strokes in accordance with the worked penetration procedure of ASTM D217, and then the mixture shall stand undisturbed for 7 days at 18 to 29.5 °C (65 to 85 °F) and 50±10 percent relative humidity. The worked penetration and then the penetration after prolonged working shall be examined in accordance with table I, using the same sample of grease mixture for both tests. Small amounts of fresh grease may be added to the worker cup just prior to the worked stability test as needed to maintain a full cup for testing.

4.5.8 Chemical and physical properties. The chemical and physical property tests shall be performed in accordance with the applicable methods of FED-STD-791, ASTM standards on petroleum and lubricants, or applicable test paragraphs as shown in table I.

4.5.8.1 Storage stability. Two standard grease worker cups (ASTM D217) shall be filled with the grease sample, one struck flush with the top of the cup for undisturbed penetration and the second filled with an additional quantity of grease to ensure sufficient sample to conduct the worked penetration. The sample for unworked penetration shall be covered with a watch glass. The sample for worked penetration shall be covered with aluminum foil. The cups shall be stored and covered in an oven at a temperature of 38±3 °C (100±5 °F) for 6 months subsequent to the original determination of the penetration. Upon termination of storage, the samples are allowed to cool to 25 °C (77 °F) prior to determining the undisturbed and worked penetration in accordance with ASTM D217. The undisturbed penetration sample shall be stored at 38±3 °C (100±5 °F) for an additional 6 months. The sample shall be covered in such a way as to provide an air space above the grease and prevent the surface of the grease from being disturbed. The sample shall then be examined for the presence of crystalline material.

4.5.8.2 Performance life test.

4.5.8.2.1 Apparatus. The apparatus shall be as shown on figure 1 and shall consist of the following:

a. Test spindle supported in a cast iron housing by two auxiliary bearings.

b. A 1-horsepower motor drives the test spindle by means of a flat 2-inch-wide belt at a speed of 3600±50 revolutions per minute.

c. Electrically heated oven with a temperature controller capable of maintaining the temperature of the test bearing within ±1.7 °C (±3 °F) of the chosen temperature. The oven shall have two separate heaters. One heater operates continuously; it is connected to a manually set variable voltage transformer for providing any voltage from 0 to 100 percent of line voltage to this heater. The second heater is connected in series with the temperature controller, thus, giving intermittent heating with this heater.

d. Test bearing: NSN 3110-00-987-3739 or single row, radial, deep groove, single shielded, size 315, bearing "NT" marking, in accordance with MIL-B-17931, heat stabilized to 149 °C (300 °F), enclosed in a housing and mounted on a vertical test spindle. A radial load of 685 newtons (154 lb) is applied to the bearing by a spring mounted outside the oven. Details of the bearing housing and its cover shall be as shown on figures 2 and 3.

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e. Temperature controller (for heater actuation) maintaining ±1.7 °C (3 °F) of set temperature.

f. Temperature recorder, to record outer race temperature of test bearing in each test unit.

g. Grease gun, hand-held, screw type with an applicator tube having a 3/32-inch minimum inside diameter.

h. Covered parts washer with heater for cleaning test bearing.

(1) Cleaning material. Electron solvent (NSN 6850-01-375-5553 [6-gallon pail], NSN 6850-01-375-5554 [1-gallon can]), or equivalent aliphatic hydrocarbon terpene mixture.

(2) Drying material. Hypersolve (NSN 6850-01-450-6162 [5-gallon pail]) or equivalent drying agent.

Note: All steps involving solvent and drying agent shall be accomplished inside a chemical fume hood with positive ventilation, while wearing solvent resistant gloves (e.g., Viton, or equal) and safety glasses with splash protection.

i. Balance, top loading, 4000-gram capacity with ±0.05-gram accuracy.

4.5.8.2.2 Preparation. For pregreased test bearings, grease shall be removed from bearings using a narrow blade nonmetallic spatula. All test bearings shall be cleaned as follows:

a. Place the bearing into a parts washer and hand wash the bearing by slowly spinning it while partially submerged in the electron solvent. The solvent shall be used at room temperature, since heating the solvent may cause the terpene to become unstable.

b. Manually remove stubborn deposits or clumps of grease with a clean brush having short, soft bristles.

c. Replace the solvent to fully cover the bearings.

d. Periodically inspect the bearing. When the visible face of the bearing is clean, rotate the bearing slightly.

e. Quickly dip the bearing in a container filled with clean solvent; quickly dip the bearing in a stainless steel container filled with drying material; place the bearing on a clean dry surface until all of the residual materials have evaporated.

f. The bearing is weighed.

g. Grease is injected into the bearing using a grease gun with a needle tip having a 3/32-inch inside diameter until 31±0.l grams of grease having a specific gravity of 0.9 are uniformly distributed throughout the bearing. Grease packs for grease with specific gravity other than 0.9 are as follows:

$$Grease\ pack\ (gram) = \frac{31\ x\ (specific\ gravity\ of\ grease)}{0.9}$$

Grease shall not extend beyond the face of the races. The bearing shield shall then be installed in the bearing housing.

h. The bearing shall be placed in the housing and then installed on the test spindle so that the bearing shield will be in the down position (see figure 1). Thermocouple wires shall be inserted in the threaded holes provided in the housings so that they are in contact with the bearing outer ring. The radial load shall then be applied.

4.5.8.2.3 Procedure. The driving motor shall be started and the bearing operated for 24 hours without heat. Maximum temperature rise over ambient temperature and temperature rise over ambient temperature at the end of 24 hours shall be noted. Heat shall then be applied to bring the bearing to the specified temperature of 149±3 °C (300±5 °F) within 1½ hours. The oven heat shall be provided principally by the variable voltage heater with the temperature controller heater acting as a trimming device to maintain the desired test bearing temperature. After 2 hours, adjust so that the heater connected to the temperature controller operates on a cycle of approximately 10 to 25 percent “on” time and 75 to 90 percent “off” time. Thereafter, no changes in heater control settings shall be made except for minor adjustments for ambient, voltage changes, and so forth. Test operation shall be cyclic with 6 days (144 hours) running followed by 1 day (24 hours) shutdown. This operating cycle shall be continued until lubrication failure occurs. During each shutdown, the outside of the bearing housing and oven shall be inspected for grease leakage. Lubrication failure shall be considered to have occurred when any of the following conditions prevail:

a. Tripping of motor thermal cut-out switch indicating that frictional torque has increased sufficiently to over-load the motor.

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b. Locking of test bearing or belt slippage (when belt tension is not a factor) on restart or during test run.

c. Excessive leakage. Leakage is excessive if grease or separated oil is slung from the bearing or housing.

d. Temperature excursion in excess of 176.7 °C (350 °F) for more than 24 hours.

e. Audible abnormal noise (such as squealing) or vibration while running or perceptible roughness when turning the test bearing by hand.

Once any of these conditions are met, the tester shall be secured and total running time recorded. The test shall be conducted on six test bearings. A Weibull plot of the results shall be used to determine the median (50 percent) life. The grease shall be reported as passing if the median life equals or exceeds that obtained in a like test using Navy Reference Grease Z-1 (see 6.7). Information on Navy Reference Grease Z-1 shall be no more than 6 months old; otherwise, it shall be run at the same time as the candidate grease.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The grease covered by this specification is intended for multipurpose use and use in grease-lubricated ball and roller bearings operating continuously in a temperature range of -29 and 149 °C (-20 and 300 °F) and up to 177 °C (350 °F) for periods up to 4 hours in any 24-hour period. This grease has unique physical and chemical requirements that are compatible for use across the Navy. Current stock of the grease may be used until depleted; future requisitions for the grease should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required.

c. When inspections should not be performed on unworked grease in accordance with the conditions specified in table I (see 4.2).

d. Packaging requirements (see 5.1).

e. Activities requiring copies of completed safety data sheets (SDS) (see 6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24508 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Provisions governing qualification. Copies of SD-6, "Provisions Governing Qualification," are available online at https://quicksearch.dla.mil/ or https://assist.dla.mil.

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6.3.2 Submission of qualification samples and additional information. When authorizing the forwarding of qualification samples, the qualifying activity will require the manufacturer to submit for review and approval, along with the samples, the manufacturer's test report (which includes the sample's SDS) (see 6.4), the location and identity of the plant which produced the sample, and the item composition report. The samples should be plainly and durably marked with the following information:

a. "Sample for Qualification Test"

b. "GREASE, HIGH PERFOMANCE, MULTIPURPOSE"

c. "Specification MIL-PRF-24508"

d. Manufacturer's name

e. Manufacturer's phone number

f. Product code number

g. Batch (see 6.8.1) or lot number

h. Date of manufacture

i. "Submitted by (name and date), for qualification inspection in accordance with MIL-PRF-24508, under authorization of (reference authorization letter)"

6.4 SDSs. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.5 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.6 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/sites/nmcphc/industrial-hygiene/pages/health-hazard-assessment.aspx.

6.7 Navy reference grease Z-1. Samples of grease required for performance life test (4.5.8.2) may be obtained by ordering products qualified to QPL 24508.

6.8 Definitions.

6.8.1 Batch. A batch is the quantity of material that has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.8.2 Lot. A lot consists of the grease produced by one manufacturer at one plant, from the same materials and under essentially the same conditions, provided the operation is continuous and does not exceed a 24-hour period.

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6.9 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, DoD Shelf-Life Management Program. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.10 Subject term (key word) listing.

High dropping point

Lubricant

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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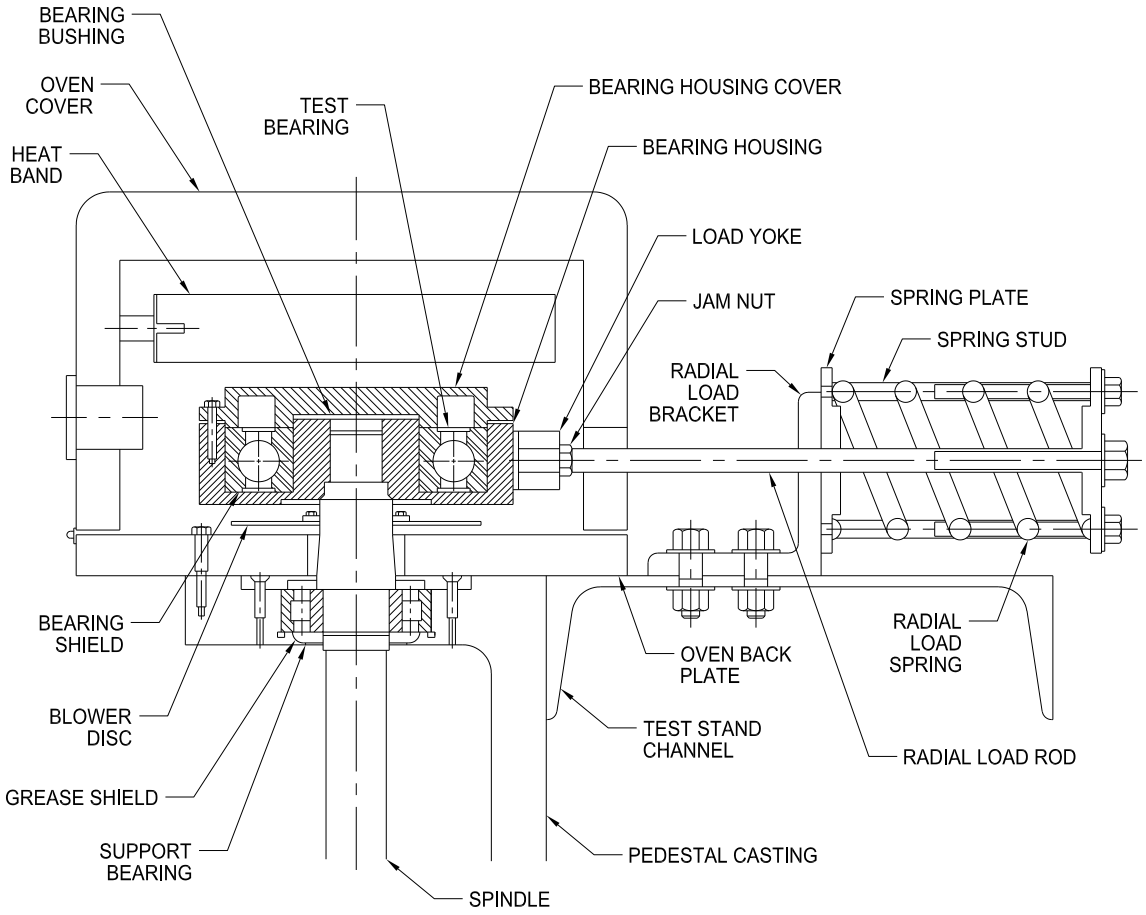


FIGURE 1. Performance tester assembly for 315 ball bearing.

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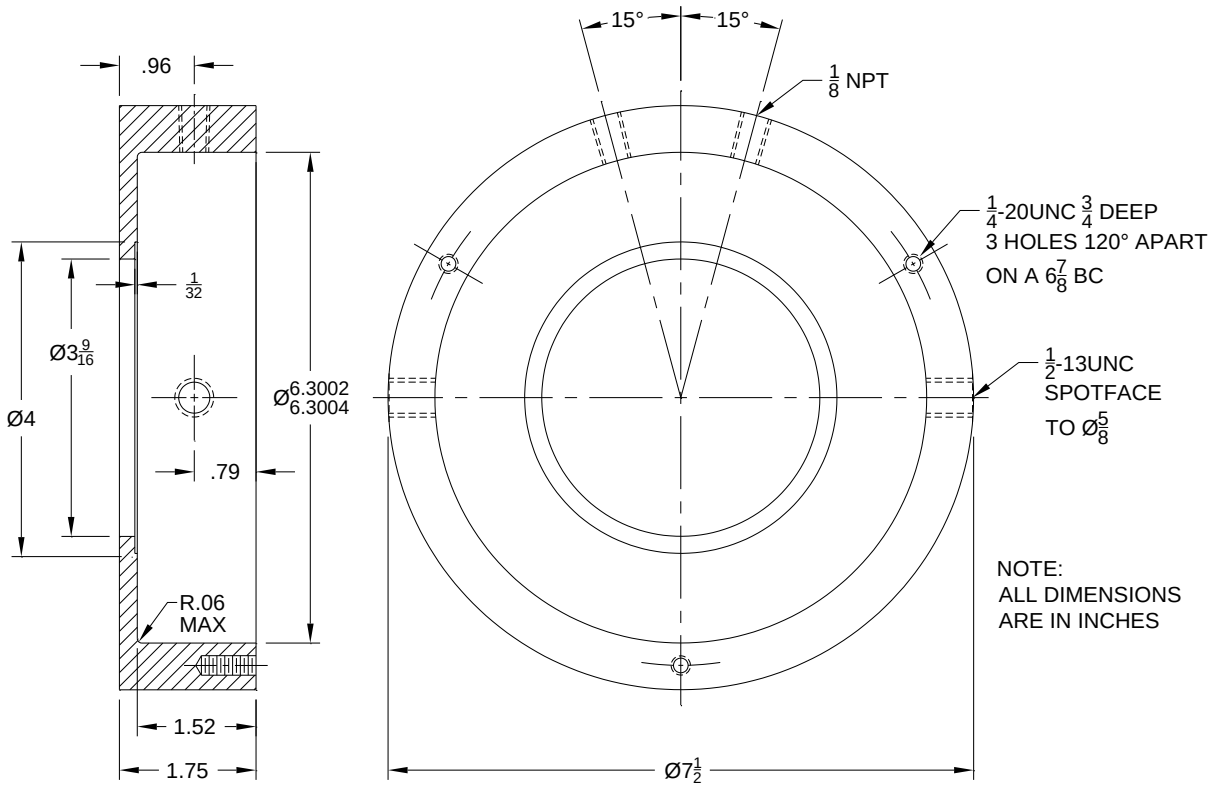


FIGURE 2. Test bearing housing.

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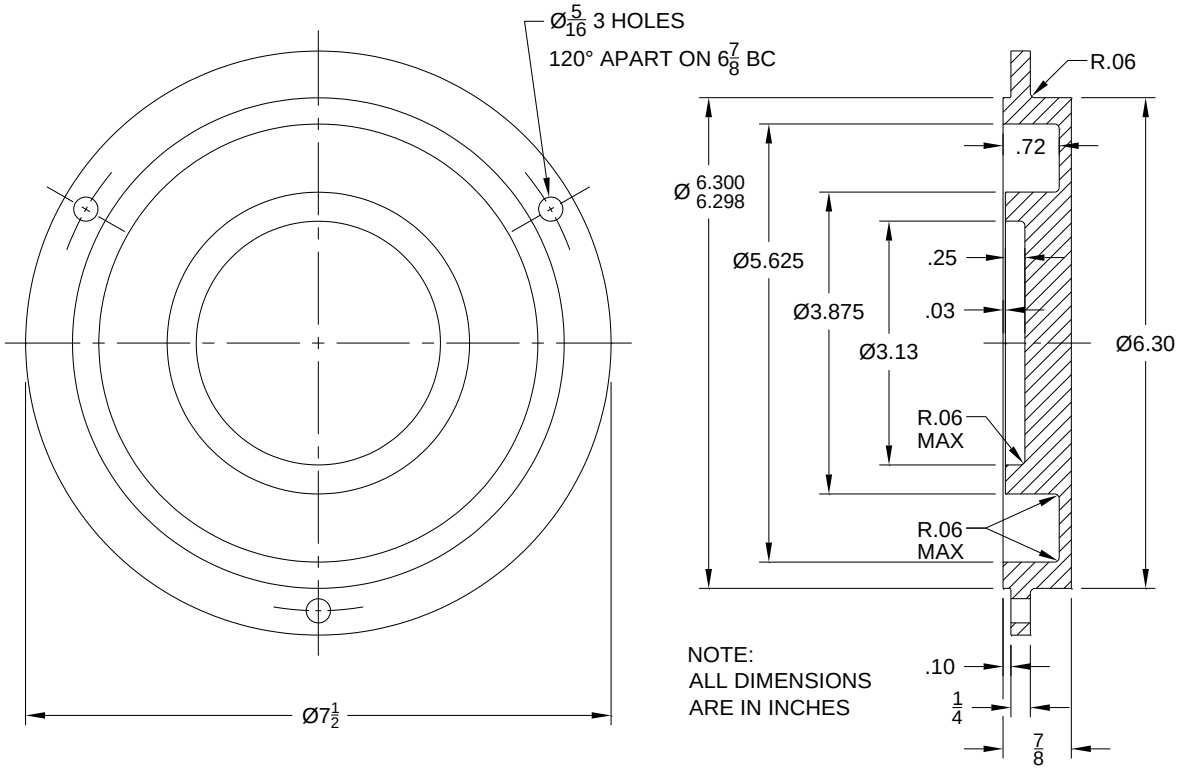


FIGURE 3. Test bearing cover.

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CONCLUDING MATERIAL

Custodians:
Navy – SH
Air Force – 68

Preparing activity:
Navy – SH
(Project 9150-2020-006)

Review activities:
Navy – AS
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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